

Use Case “Add/Update Product (Product Manager)”

1. Use case code

UC001

2. Brief Description

This use case describes the interaction between the Product Manager and the AIMS system when the Product Manager wishes to add a new product or update an existing product.

3. Actors

3.1. Product Manager

4. Preconditions

- The Product Manager is logged in with the appropriate role to access product-maintenance features.
- For Update: the target product already exists in the catalog.

5. Basic Flow of Events (Success)

1. Product Manager requests to Add a new product
2. Software displays the product form with a product type selector (Book, Newspaper, CD, DVD).
3. Product Manager enters product data, including title/category (AIMS supports search by product title or category), original value, current price, and type-specific fields as required.
4. Software validates:
 - Price policy: current price is within 30%–150% of the original value.

- Type-specific required fields are present (per product type list below).
- Date formats are valid where applicable (e.g., publication date). If invalid, the system notifies the Product Manager.

5. Software performs operation Add by creating a new product record.

6. Software stores a history entry for the addition or edit.

7. Software confirms success and shows the saved product summary.

6. Alternative Flows

Table 2 — Alternative flows of events for UC “Add/Update Product”

No	Location	Condition	Action	Resume location
1	At Step 1	User chooses Update operation	Software displays Update form	Step 2
2	At Step 4	Price violates the 30%–150% band	Software notifies invalid price; submission rejected	Step 2
3	At Step 4	Missing type-specific required info	Software notifies invalid/insufficient info	Step 2
4	At Step 4	Invalid date format (e.g., publication date)	Software notifies invalid format	Step 2
5	At Step 5	User performs Update operation	Software performs operation Update: applies the submitted changes to the existing record	Step 6
6	At Step 3 (Update)	Product Manager adjusts stock quantity manually	Software requires reason for stock change and records it	Step 4

7. Input data

Table A — Input data for Add/Update Product

A.1 Common fields (all products)

No	Data fields	Description	Mandatory	Valid condition / Notes
1	Title	Title of products	Yes	Must be between 3 and 100 characters long; no special characters other than hyphen and apostrophe.
2	Category / Product type	(Book, Newspaper, CD, DVD)	Yes	One of the supported product types
3	Original value	Implicitly inferred from the requirements	Yes	> 0; used for price-band validation (see A.6);
4	Current price	The current price after adjustments from the original price	Yes	Must be within 30%–150% of original value
5	Stock quantity (if updating quantity)		Conditional (Update)	Manual adjustments must record a reason

A.2 Book-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Author(s)	Book authors	Yes	Must be between 1 and 50 characters; can contain letters, spaces, and commas.	“Robert C. Martin”
2	Cover type	Paperback or hardcover	Yes	One of {Paperback, Hardcover}	Hardcover
3	Publisher	Publishing house	Yes	Must be between 3 and 100 characters; only alphabetic characters, spaces, and hyphens.	Pearson

4	Publication date	Date of publication	Yes	Must follow the format dd/mm/yyyy; the date must be a valid date (e.g., 30/12/2018).	30/12/2018
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A.3 Newspaper-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Editor-in-chief	Lead editor	Yes	Must be between 3 and 100 characters long; only alphabetic characters, spaces, and hyphens.	Nguyen Van A
2	Publisher	Publishing organization	Yes	Must be between 3 and 100 characters; only alphabetic characters, spaces, and hyphens.	Vietnam News Agency
3	Publication date	Date of issue	Yes	Must follow the format dd/mm/yyyy; the date must be a valid date (e.g., 05/10/2025).	05/10/2025
4	Issue number	Issue identifier	No	Must be a positive integer (e.g., 342).	342
5	Publication frequency	Daily/Weekly/etc.	No	Must be one of {Daily, Weekly, Monthly}.	Weekly
6	ISSN	International Standard Serial Number	No	Must be 8 digits, with a hyphen between each 4 digits.	0868-1234
7	Language	Publication language	No	Must be a valid language.	Vietnamese
8	Sections	e.g., politics, business, sports, culture	No	Must be a valid section from section rules.	Politics; Business

A.4 CD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Artists	Performer(s)	Yes	Must be between 3 and 100 characters long; only alphabetic characters, spaces, and hyphens.	"Adele"
2	Record label	Label/publisher	Yes	Must be between 3 and 100 characters long; only alphabetic characters, spaces, and hyphens.	XL Recordings
3	Tracks list	List of tracks	Yes	Each track has title and length	"Track 01 – Hello – 4:55"
4	Genre	Musical genre	Yes	Must be a valid musical genre from musical genre rules.	Pop
5	Release date	Date of release	No	Must follow the format dd/mm/yyyy; the date must be a valid date (e.g., 05/10/2025).	20/11/2015

A.5 DVD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Disc type	Blu-ray or HD-DVD	Yes	One of {Blu-ray, HD-DVD}	Blu-ray
2	Director	Film director	Yes	Must be between 3 and 100 characters long; only alphabetic characters, spaces, and hyphens.	Christopher Nolan
3	Runtime	Total running time	Yes	Positive duration	152 min
4	Studio	Production studio	Yes	Must be between 3 and 100 characters long; only alphabetic characters, spaces, and hyphens.	Warner Bros.

5	Language	Audio language(s)	Yes	Must be a valid language.	English
6	Subtitles	Subtitle language(s)	Yes	Must be valid languages.	English; Vietnamese
7	Release date	Date of release	No	Must follow the format dd/mm/yyyy; the date must be a valid date (e.g., 05/10/2025).	16/07/2010
8	Genre	Film genre	No	Must be a valid genre from film genre rules.	Action

Validation note: When product data **does not comply with rules** or **date format is incorrect**, the software **notifies** the Product Manager.

8. Output data

Table B — Output data after successful Add/Update

No	Data fields	Description
1	Product title & type	The saved product and its type (Book/Newspaper/CD/DVD)
2	Original value & current price	Confirmed values satisfying the 30%–150% band
3	Type-specific attributes	The saved attributes according to the product type (see A.2–A.5)
4	History entry recorded	Addition/edit stored in history

9. Postconditions

- Add: A new product exists in the catalog; a history entry is recorded.
- Update: The product reflects submitted changes; a history entry is recorded.
- Saved current price respects the 30%–150% policy; required type-specific attributes are present.

